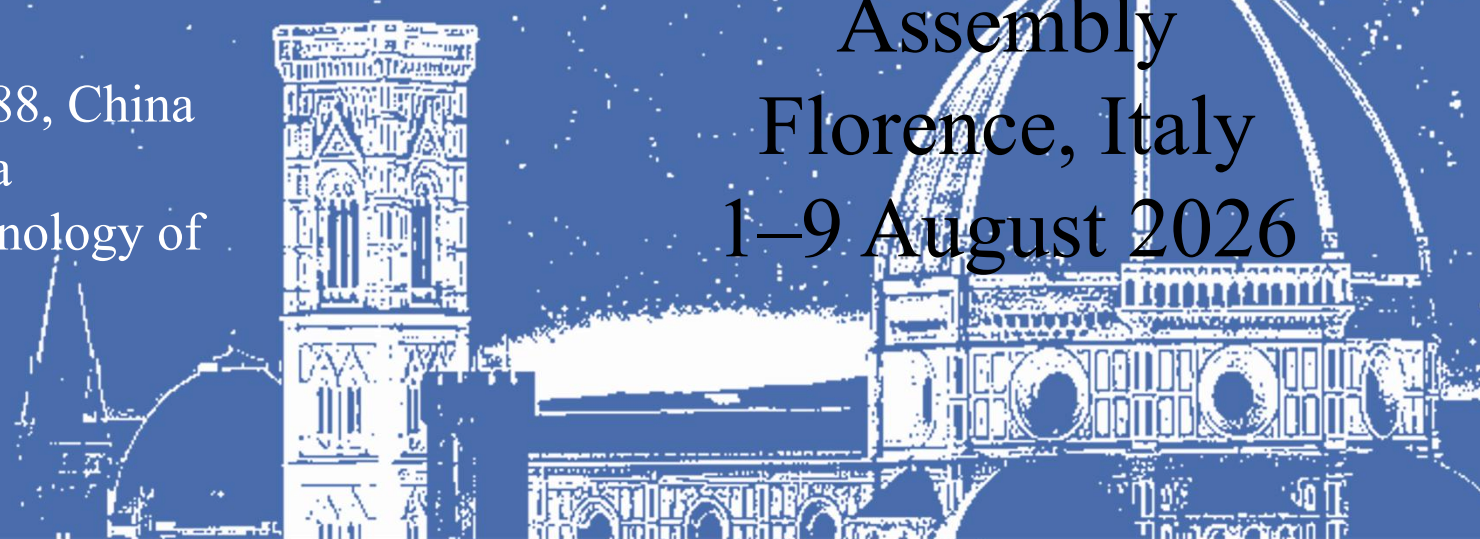
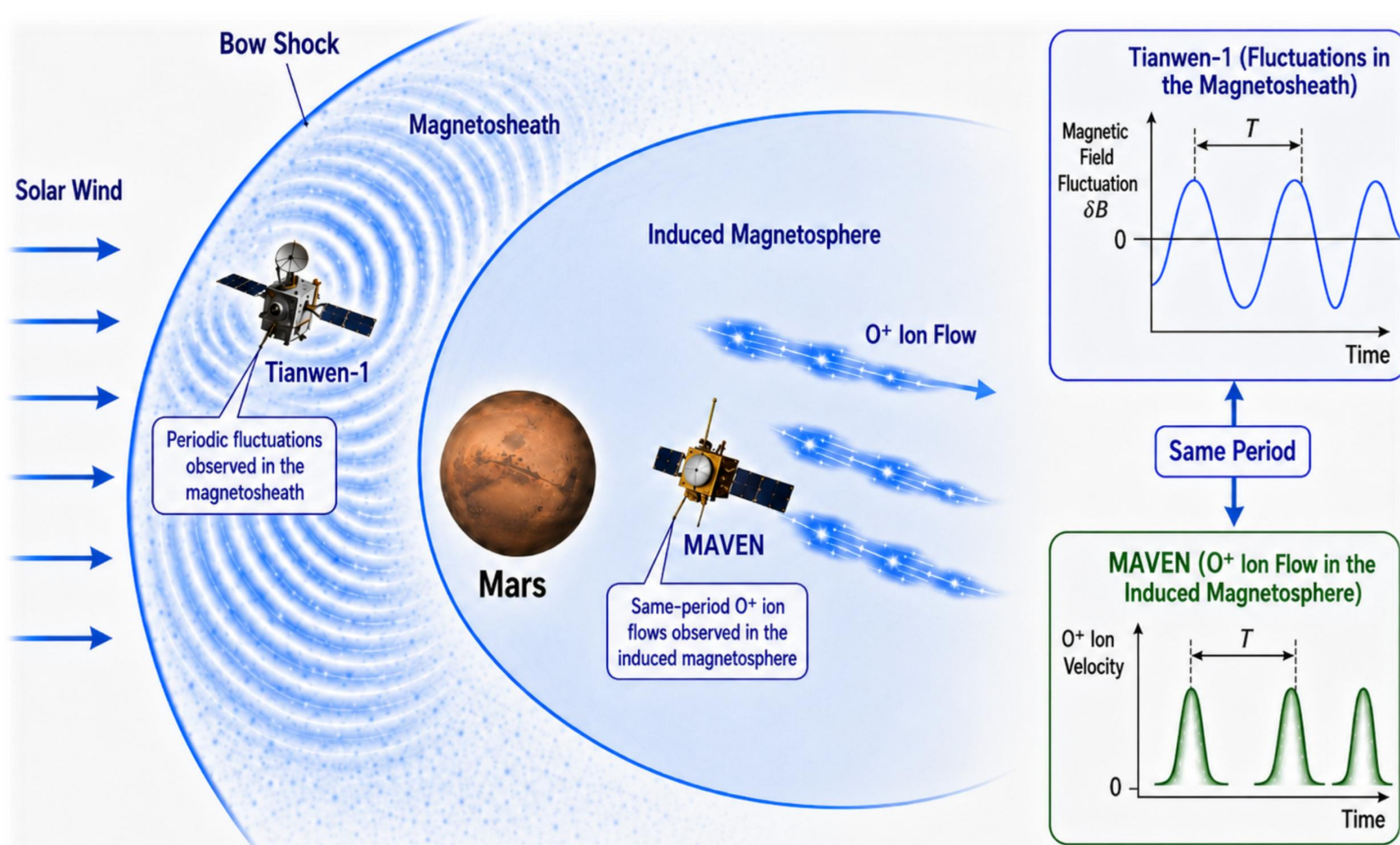




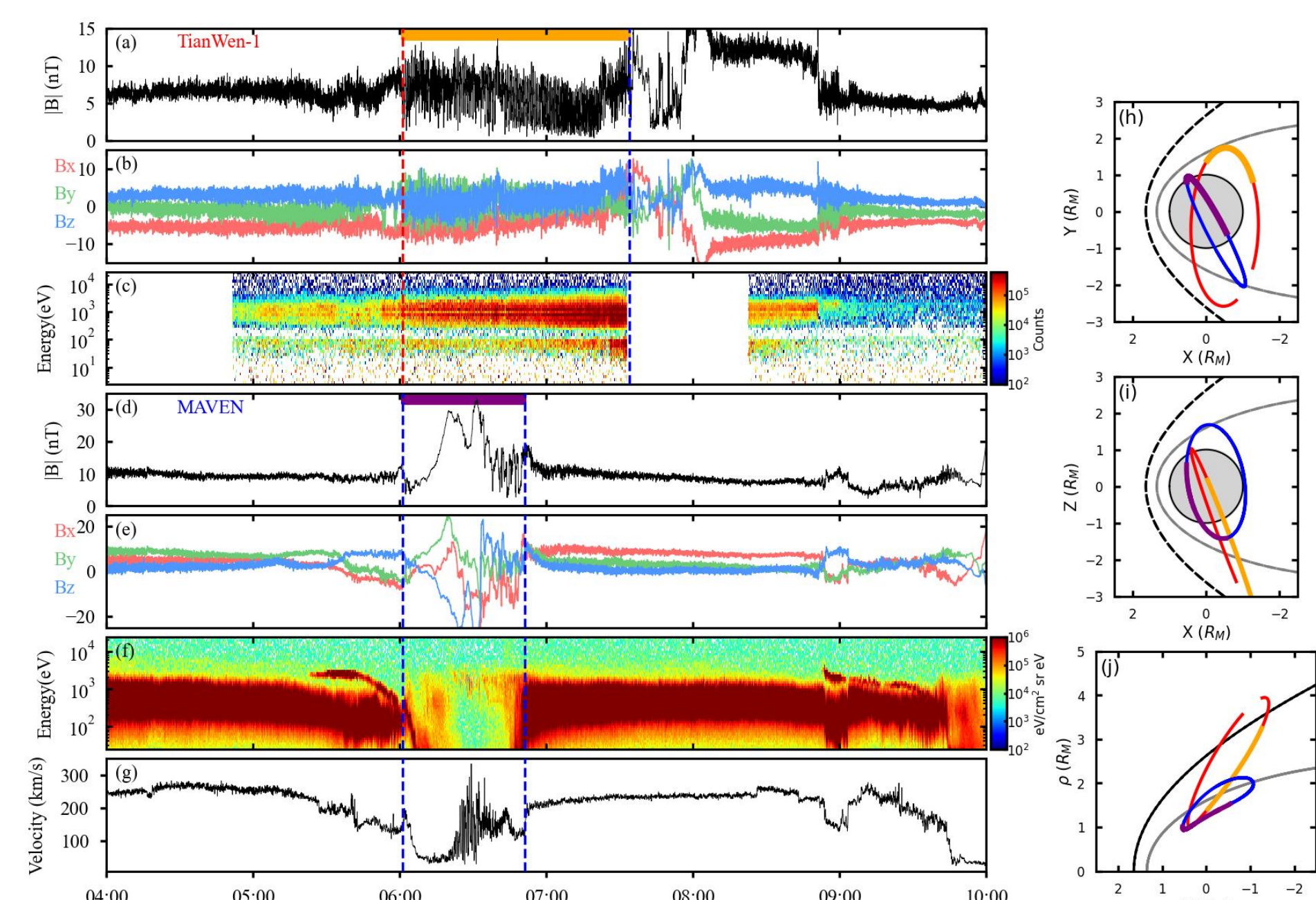
Abstract



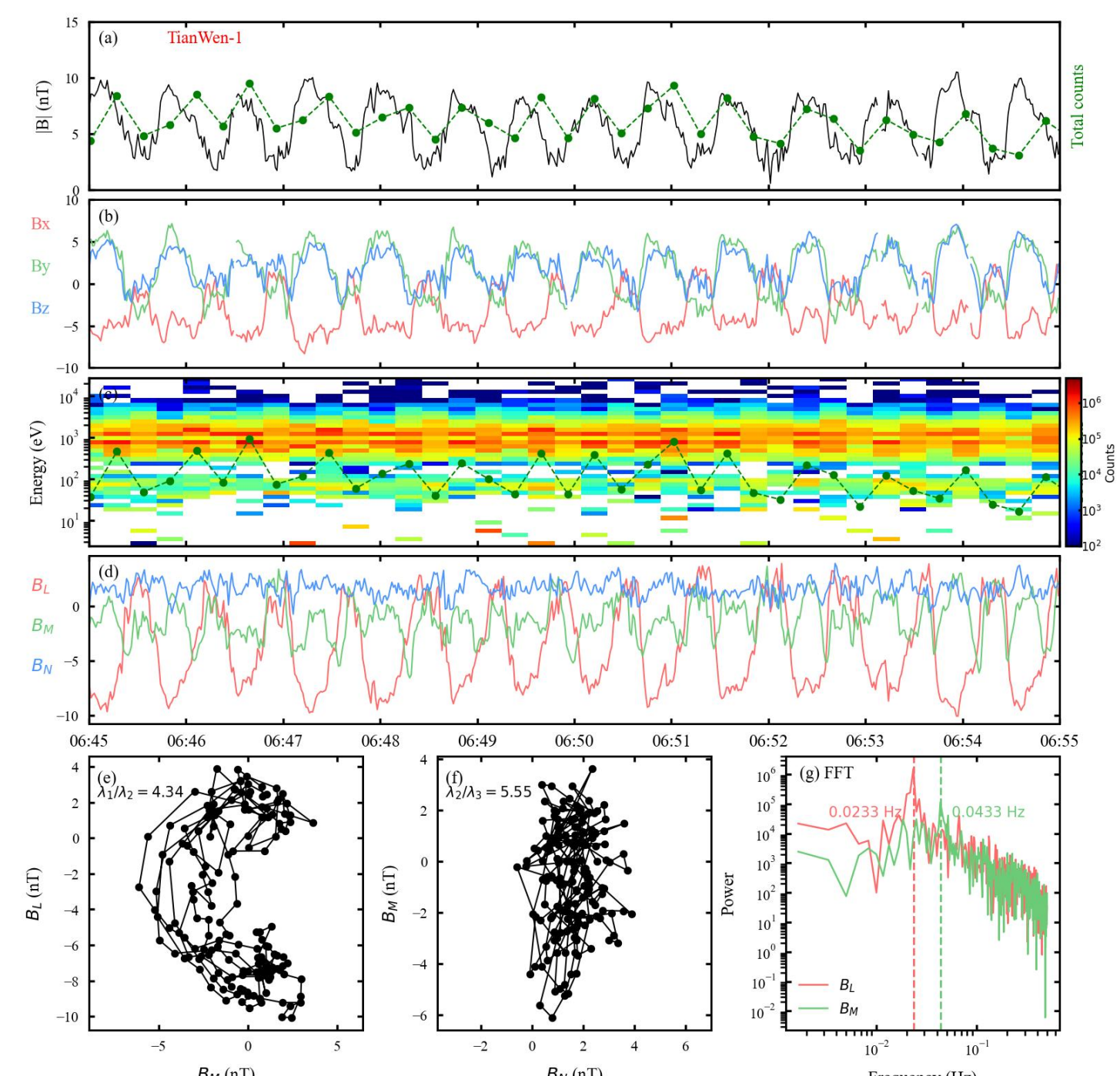
Mars is currently the only planet besides Earth where its space environment can be investigated through coordinated multi-spacecraft observations. Here, using magnetic field (MOMAG) and plasma (MINPA) measurements from Tianwen-1, we found fast-mode waves in the Martian magnetosheath with a period of approximately 50 s. Meanwhile, MAVEN, located in the Martian induced magnetosphere, observes ion flows with the same period, predominantly composed of O⁺ ions. The coordinated observations from the two spacecraft indicate that periodic fast mode waves in the Martian magnetosheath can significantly modulate O⁺ ion flows in the induced magnetosphere.

Result

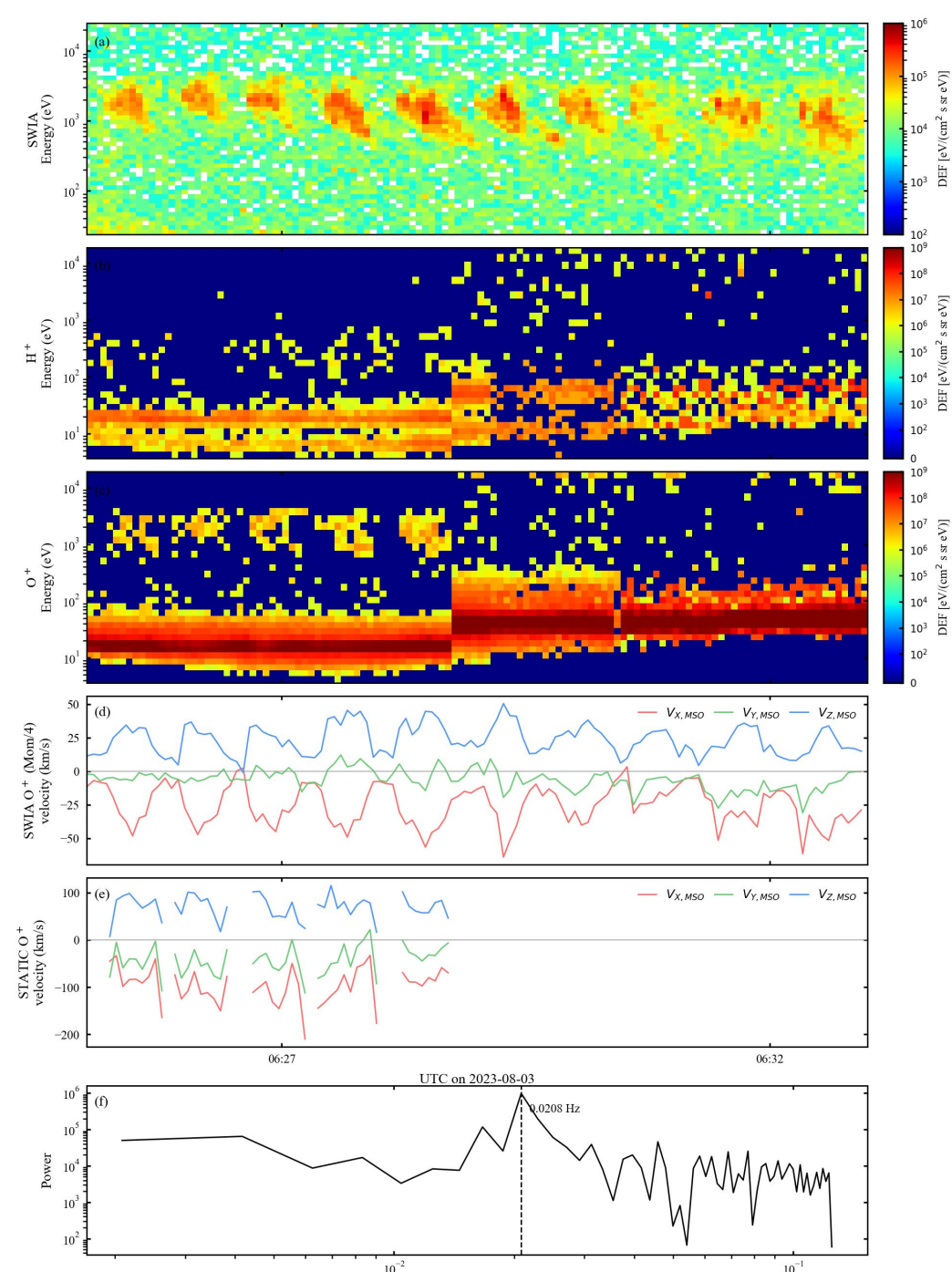
(a) Simultaneous observations by Tianwen-1 and MAVEN on August 3, 2023.



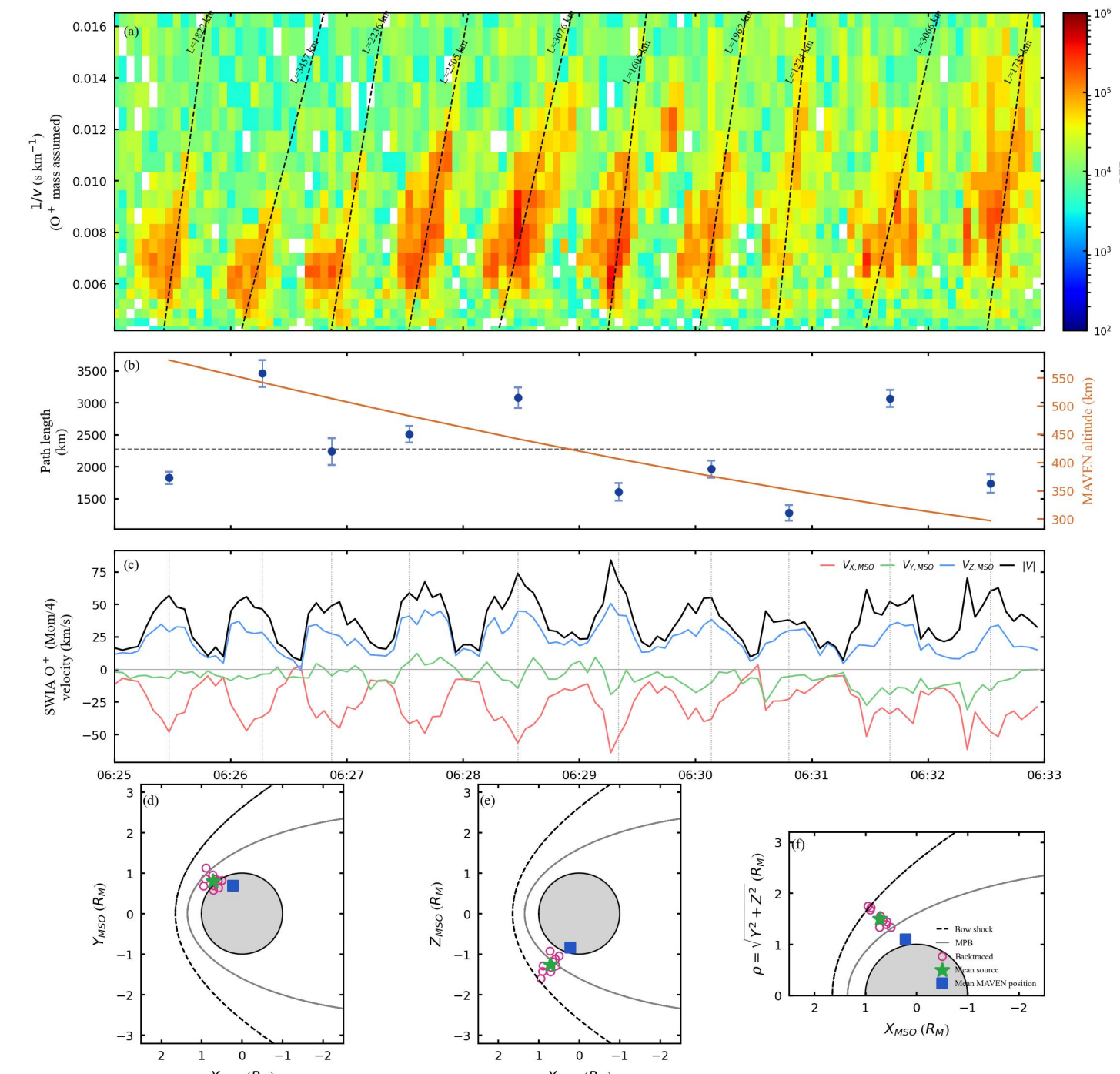
(b) Tianwen-1 observed ~50 s fast-mode waves in magnetosheath.



(c) ~50 s Periodic O⁺ Ion Flows Observed by MAVEN in the Induced Magnetosphere



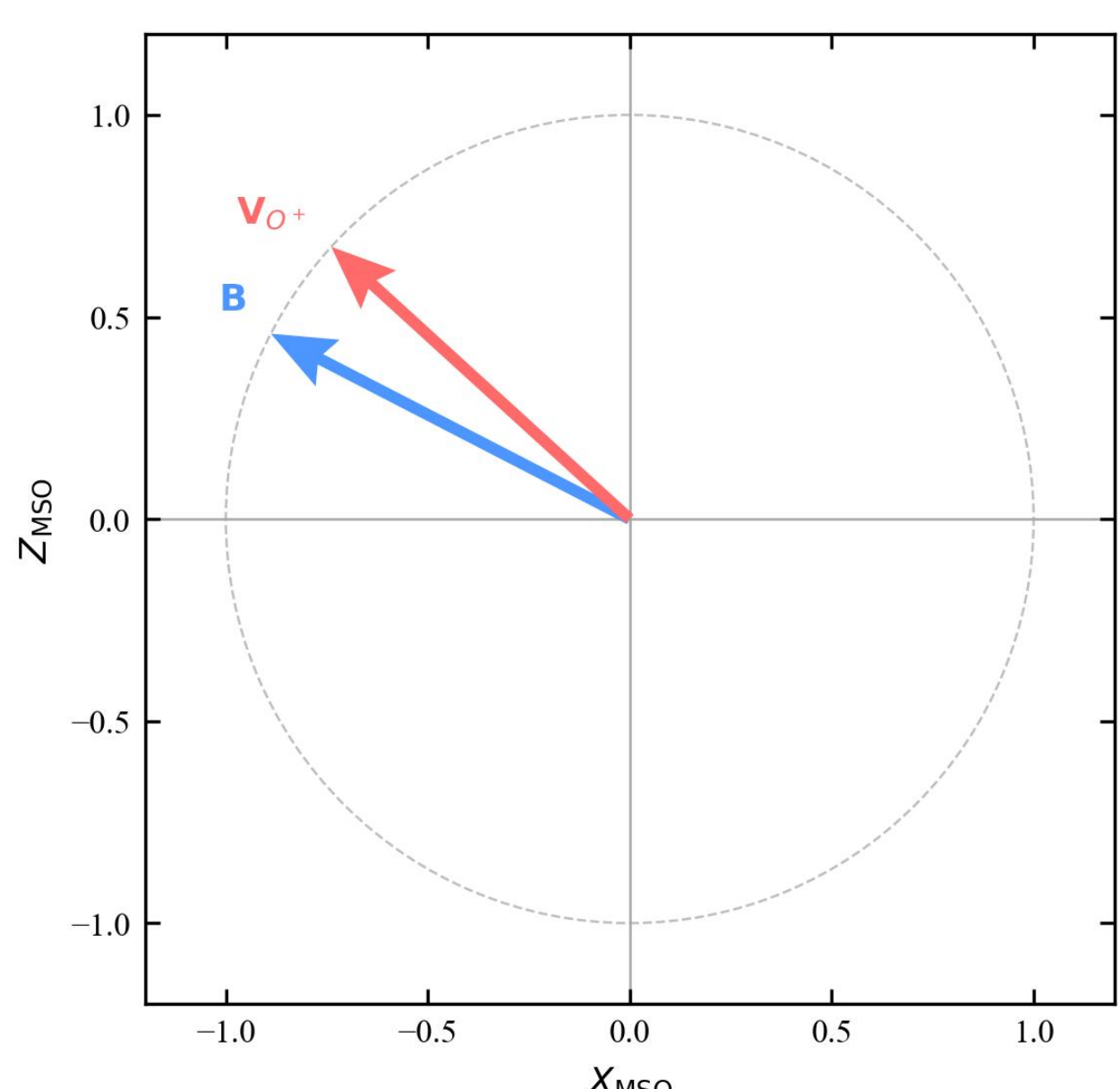
(d) Infer the O⁺ source location using velocity dispersion.



► Tianwen-1 observed a significant enhancement in both the magnetic field fluctuations and ion counts, indicating that the spacecraft had entered the Martian magnetosheath, while MAVEN was located within the induced magnetosphere (Figure a). In the magnetosheath, the Tianwen-1 magnetometer (MOMAG) detected magnetic field fluctuations with a period of approximately 50 s, characterized by strong compressibility. Simultaneously, the MINPA instrument observed periodic enhancements in the total ion counts that were synchronized with the magnetic field fluctuations. In addition, a clear second harmonic was identified. The existence of fast-mode waves and their second harmonics in the Martian magnetosheath was first reported by Bharati Kakad et al. (2024). These observational signatures provide strong evidence that the waves observed by Tianwen-1 are fast-mode waves (Figure b). At the same time, MAVEN, located within the induced magnetosphere, observed periodic ion flows with a similar period of approximately 50 s. Measurements from STATIC indicate that these periodic ion flows were dominated by O⁺ ions (Figure c), and their velocity vectors were primarily confined to the XZ plane. Using the time-of-flight relation, $t=t_0+L/v$, together with the observed O⁺ ions velocities, we infer that these periodic O⁺ ion flows originated from the high-altitude magnetosheath, suggesting that their generation is likely associated with the magnetosheath waves observed by Tianwen-1 (Figure d).

Discussion

(e) Magnetosheath Magnetic Field and O⁺ Ion Velocity in the XZ Plane



At present, two mechanisms have been proposed to explain the generation of periodic ion flows in the Martian induced magnetosphere. One attributes them to a periodically varying solar wind electric field (Halekas et al., 2015), while the other suggests that they are generated by magnetic reconnection (Zhang et al., 2021). If these periodic O⁺ ion flows were driven by the convective electric field, $E=-V \times B$, their velocities would be expected to be nearly perpendicular to the magnetosheath magnetic field. However, the calculated angle between the magnetosheath magnetic field and the O⁺ ion velocity is only 34.8°. The O⁺ ion velocity is primarily confined to the XZ plane, and the XZ-plane component of the magnetosheath magnetic field is nearly parallel to the O⁺ ion velocity (Figure e). Therefore, we suggest that these periodic O⁺ ion flows were likely generated by magnetic reconnection occurring in the XZ plane.

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Acknowledgments

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Poster PDF & Contact



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